Macroeconomics PhD Intro
Lecture 3: Dynamic Optimization Methods

Chapter 4: Sequential and Recursive Methods

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Learning Goals

- Formulate dynamic optimization problems with states, controls, and feasibility sets.
- Solve finite-horizon models with Kuhn-Tucker conditions and Euler equations.
- Understand infinite-horizon subtleties: boundedness, no-Ponzi condition, and transversality.
- Derive and interpret Bellman equations and policy functions.
- Connect sequential and recursive methods in the neoclassical growth model.
- Build intuition for convergence and balanced growth under optimizing behavior.

Road Map (Following Chapter 4)

1. Dynamic optimization setup and two benchmark models.
2. Sequential methods with finite horizon.
3. Sequential methods with infinite horizon.
4. Recursive methods and the Bellman equation.
5. Dynamics and convergence in the optimizing neoclassical growth model.

Connection to Lecture 2

In Solow, saving was exogenous. In this lecture, saving is chosen optimally from primitives.

Why Dynamic Optimization?

Cass-Koopmans adds micro-foundations to growth: saving/investment become outcomes of rational choice.

- Solow: $i_t = sy_t$ imposed.
- Optimizing growth: i_t implied by Euler equation.

What Changes?

- Positive analysis: richer transition dynamics and policy propagation.
- Normative analysis: welfare comparisons become internally coherent.

Sequential vs Recursive: Conceptual Difference

Sequential

- Unknown: entire sequence.
- Method: Kuhn-Tucker/FOCs over time.
- Output: path $\{x_t, y_t\}$.

Recursive

- Unknown: value function $V(\cdot)$.
- Method: functional equation.
- Output: policy rule $x' = g(x)$.

Both represent the same economics under standard assumptions.

Generic Dynamic Optimization Problem (P1)

$$\max_{\{y_t, x_{t+1}\}_{t=0}^{T}} \sum_{t=0}^{T} \beta^t \hat{\mathcal{F}}(y_t)$$

subject to

$$x_{t+1} = h(x_t, y_t), \qquad x_{t+1} \in \Gamma(x_t)$$

- State: x_t (predetermined stock variable).
- Control: y_t (choice in period t).
- Initial condition: x_0 given.

Appendix

Discounting and Intertemporal Trade-offs

- $\beta \in (0, 1)$: stationary discount factor.
- Relative welfare weight between dates t and $t + k$: β^k.
- Dynamic trade-off: reducing current payoff can raise future payoff through state dynamics.

Interpretation

The model separates **technology/constraints** $\{h, \Gamma\}$ from **preferences/objective** $\{\hat{\mathcal{F}}, \beta\}$.

Existence Intuition

Sufficient conditions used in Chapter 4:

- Continuity of objective and transition mappings.
- Feasible set nonempty, closed, bounded.
- Concavity/convexity structure.

Why it matters:

- A maximum exists (not just a supremum).
- FOCs characterize a unique global solution when problem is convex.

State-Control Timeline

Given x_t, choosing y_t determines $x_{t+1} = h(x_t, y_t)$.

- At $t = 0$: x_0 exogenous, choose y_0, get x_1.
- At $t = 1$: x_1 now predetermined, choose y_1, get x_2.
- Continue recursively.

Key Distinction

Controls are current choices. States are consequences of past choices.

Consumption-Saving Model: Primitives

- Preferences: $U = \sum_{t=0}^{T} \beta^t u(c_t)$.
- $u'(c) > 0$, $u''(c) < 0$, and $\lim_{c \to 0} u'(c) = \infty$.
- Endowment income: wage sequence $\{w_t\}$.
- Assets a_t with return r_t.

Budget constraint:

$$c_t = w_t + (1 + r_t)a_t - a_{t+1}$$

Borrowing limit and terminal condition:

$$a_{t+1} \geq \underline{a}, \qquad a_{T+1} \geq 0$$

Economic Interpretation of Asset Dynamics

- If $a_{t+1} > a_t(1 + r_t) + w_t$, consumption is negative (infeasible).
- $a_{t+1} < 0$ means borrowing from the future.
- Terminal restriction avoids "borrow and default" in the last period.

Trade-off

One unit less consumption today buys $1 + r_{t+1}$ more resources tomorrow.

Consumption-Saving Problem (P2)

$$\max_{\{c_t, a_{t+1}\}_{t=0}^{T}} \sum_{t=0}^{T} \beta^t u(c_t)$$

subject to

$$c_t = w_t + (1 + r_t)a_t - a_{t+1}$$

$$c_t \geq 0, \quad a_{t+1} \geq \underline{a}, \quad a_{T+1} \geq 0$$

Mapping to (P1): state $x_t = a_t$, control $y_t = c_t$.

Neoclassical Growth Model: Setup

Production economy (not pure endowment):

$$y_t = f(k_t), \qquad k_{t+1} = (1-\delta)k_t + i_t$$

Resource constraint:

$$c_t + i_t \leq y_t$$

Equivalent form:

$$c_t + k_{t+1} \leq f(k_t) + (1-\delta)k_t$$

Same preferences as consumption-saving model, but return is endogenous through $f'(k)$.

Planner Problem in the NGM (4.3)

$$\max_{\{c_t, k_{t+1}\}_{t=0}^{T}} \sum_{t=0}^{T} \beta^t u(c_t)$$

subject to

$$c_t + k_{t+1} \leq f(k_t) + (1-\delta)k_t, \qquad c_t, k_{t+1} \geq 0$$

Interpretation

Competitive equilibrium with complete markets and perfect competition will decentralize this planner allocation (Chapter 5-6 logic).

Finite Horizon: Strategy

When $T < \infty$, use Kuhn-Tucker conditions.

Why this is tractable:

- Finite number of unknowns.
- Boundary condition at the terminal date.
- Concavity gives sufficiency of FOCs.

Two-Period Consumption-Saving Example

With $a_0 = 0$, $\underline{a} = 0$, and constant r:

$$c_0 = w_0 - a_1, \qquad c_1 = w_1 + (1 + r)a_1$$

Lifetime budget:

$$c_0 + \frac{c_1}{1 + r} = w_0 + \frac{w_1}{1 + r}$$

Borrowing/saving restriction: $a_1 \geq 0 \Leftrightarrow w_0 - c_0 \geq 0$.

Appendix

Two-Period Lagrangian and FOCs

Lagrangian:

$$\mathcal{L} = u(c_0) + \beta u(c_1) + \mu \left(w_0 + \frac{w_1}{1+r} - c_0 - \frac{c_1}{1+r} \right) + \lambda(w_0 - c_0)$$

FOCs:

$$u'(c_0) - \mu - \lambda = 0, \qquad \beta u'(c_1) - \frac{\mu}{1+r} = 0$$

Complementary slackness:

$$\lambda(w_0 - c_0) = 0, \quad w_0 - c_0 \geq 0, \quad \lambda \geq 0$$

Euler Equation and Cases

If interior ($\lambda = 0$):

$$u'(c_0) = \beta(1+r)u'(c_1)$$

- $\beta(1+r) = 1$: strongest smoothing benchmark.
- High current income and borrowing constraint can generate corner solutions.
- If borrowing bound binds, Euler inequality replaces equality.

Generic T-Period Result (Result 1)

Rewrite objective as

$$\max_{\{x_{t+1}\}} \sum_{t=0}^{T} \beta^t \mathcal{F}(x_t, x_{t+1}), \qquad x_{t+1} \in [\underline{x}, \gamma(x_t)]$$

Interior Euler equation for $t < T$:

$$\mathcal{F}_2(x_t^*, x_{t+1}^*) + \beta \mathcal{F}_1(x_{t+1}^*, x_{t+2}^*) = 0$$

Terminal condition:

$$x_{T+1}^* = \underline{x}$$

This is a second-order difference system in the state sequence.

Finite-Horizon NGM in Sequence Form (P3)

$$\max_{\{k_{t+1}\}_{t=0}^{T}} \sum_{t=0}^{T} \beta^t u\left(f(k_t) + (1-\delta)k_t - k_{t+1}\right)$$

subject to

$$k_{t+1} \geq 0$$

FOC for $t < T$:

$$-u'(c_t) + \beta u'(c_{t+1})\left[f'(k_{t+1}) + 1 - \delta\right] + \mu_t = 0$$

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Terminal Condition in Finite-Horizon NGM

Final-period FOC:

$$-u'(c_T) + \mu_T = 0$$

Because $u'(c_T) > 0$, we get $\mu_T > 0$, hence by complementary slackness:

$$k_{T+1} = 0$$

Intuition

Capital after the terminal date has no continuation value, so it is never optimal to leave it unconsumed.

Euler Equation for NGM (Interior Dates)

For $t < T$, Inada conditions imply interior $k_{t+1} > 0$ and $\mu_t = 0$:

$$u'(c_t) = \beta u'(c_{t+1}) \left[f'(k_{t+1}) + 1 - \delta \right]$$

with

$$c_t = f(k_t) + (1 - \delta) k_t - k_{t+1}$$

Unknowns pinned down by: Euler equations + k_0 + terminal condition $k_{T+1} = 0$.

Example 4.1: Closed-Form Finite-Horizon Solution

Assume:

$$u(c) = \log c, \qquad f(k) = Ak^{\alpha}, \qquad \delta = 1$$

Then

$$k_{t+1} = \alpha\beta \frac{1 - (\alpha\beta)^{T-t}}{1 - (\alpha\beta)^{T-t+1}} Ak_t^{\alpha}$$

and

$$c_t = \frac{1 - \alpha\beta}{1 - (\alpha\beta)^{T-t+1}} Ak_t^{\alpha}$$

Message: optimal saving rate is time-varying because horizon is finite.

Finite-Horizon Intuition: Time-to-Go Effect

- Far from terminal date: strong incentive to invest for future periods.
- Near terminal date: investment motive weakens.
- Last period: invest nothing beyond horizon.

Contrast with Infinite Horizon

In stationary infinite-horizon models, rules are time-invariant and depend on the state, not calendar time.

CRRA Utility Class

$$u(c) = \frac{c^{1-\sigma} - 1}{1 - \sigma}, \qquad \sigma \geq 0,\ \sigma \neq 1$$

Special cases:

- $\sigma \to 1$: log utility.
- $\sigma > 0$: strictly concave preferences.
- $\sigma = 0$: linear utility.

CRRA is central because it delivers constant EIS and is compatible with balanced growth.

Example 4.2: EIS Under CRRA

With linear returns $R_{t,t+s}$, repeated Euler substitution gives

$$\frac{c_{t+s}}{c_t} = (\beta^s R_{t,t+s})^{1/\sigma}$$

Hence

$$EIS = \frac{\partial \log(c_{t+s}/c_t)}{\partial \log R_{t,t+s}} = \frac{1}{\sigma}$$

- High σ: lower willingness to tilt consumption intertemporally.
- Low σ: stronger response to return changes.

Appendix

Why Infinite Horizon?

- Stationarity: problem at t has same structure as at $t + 1$.
- Useful approximation for long-run macro questions.
- Dynastic interpretation: current generation values descendants.

But infinite horizon requires extra mathematical discipline.

Infinite-Horizon Pitfall 1: Unbounded Utility

Even with $\beta < 1$, utility may be ill-defined if feasible consumption grows too fast.
For CRRA with growth $c_t = c_0(1+\gamma)^t$, boundedness needs:

$$\beta(1+\gamma)^{1-\sigma} < 1$$

Takeaway

Discounting alone is not enough; curvature and growth jointly determine boundedness.

Infinite-Horizon Pitfall 2: Too-Large Feasible Set

Without borrowing limits, households can increase c_0 by rolling debt forever.

Need no-Ponzi condition (nPg):

$$\lim_{t \to \infty} \frac{a_{t+1}}{(1+r)^t} \geq 0$$

- Excludes exploding debt in present value.
- Makes optimization problem well posed.

nPg and Lifetime Budget Constraint

From period-by-period budget constraints:

$$c_t + a_{t+1} = (1 + r)a_t$$

Iterating forward and using nPg gives

$$\sum_{t=0}^{\infty} \frac{c_t}{(1 + r)^t} \leq a_0(1 + r)$$

At optimum with strictly increasing utility, equality holds.

Appendix

TVC: Distinct from nPg

Transversality condition:

$$\lim_{t\to\infty} \frac{a_{t+1}}{(1+r)^t} \leq 0$$

- nPg: external feasibility restriction (no endless unpaid debt).
- TVC: internal optimality restriction (no over-accumulating assets that could be consumed).

Together with strict monotonicity of utility, they imply a tight lifetime budget equality.

Example 4.3: Infinite-Horizon Log Consumption-Saving

Problem:

$$\max \sum_{t=0}^{\infty} \beta^t \log c_t \quad \text{s.t.} \quad c_t + a_{t+1} = (1+r)a_t$$

Consolidated budget and FOCs imply:

$$c_t = [\beta(1+r)]^t c_0, \qquad c_0 = a_0(1-\beta)(1+r)$$

This is a full closed-form consumption path.

Proposition 4.4 (Sufficiency with TVC)

For

$$\max_{\{x_{t+1}\}} \sum_{t=0}^{\infty} \beta^t \mathcal{F}(x_t, x_{t+1}), \quad x_{t+1} \in \Gamma(x_t)$$

if

- Euler equations hold at all t,
- TVC holds,
- $\mathcal{F}$ jointly concave and increasing in first argument,
- Γ convex,

then the candidate sequence is optimal.

Appendix

TVC in the Infinite-Horizon NGM

In NGM:

$$\mathcal{F}_1 \to u'(c_t)\left[f'(k_t) + 1 - \delta\right]$$

TVC becomes

$$\lim_{t\to\infty} \beta^t u'(c_t)\left[f'(k_t) + 1 - \delta\right] k_t = 0$$

Economic Meaning

The discounted shadow value of remaining capital cannot stay positive in the limit.

Infinite-Horizon NGM: Steady State Condition

At a stationary steady state ($c_{t+1} = c_t$, $k_{t+1} = k_t = \bar{k}$), Euler implies

$$1 = \beta \left[f'(\bar{k}) + 1 - \delta \right]$$

Then

$$\bar{c} = f(\bar{k}) - \delta \bar{k}$$

This is the optimizing counterpart to Solow steady-state logic.

Examples 4.5-4.6: Link to Solow

For log utility, Cobb-Douglas, full depreciation:

$$k_{t+1} = \alpha\beta A k_t^{\alpha}$$

Hence saving is a constant share:

$$s = \alpha\beta$$

Important Bridge

Under these assumptions, optimizing growth generates Solow-style law of motion with endogenous s.

Balanced Growth Version of NGM

With $Y_t = F(K_t, A_t L_t)$, A_t grows at γ, L_t at n, and utilitarian welfare:

$$\max \sum_{t=0}^{\infty} \beta^t L_t u(c_t)$$

Normalize by efficiency units:

$$\tilde{k}_t = \frac{K_t}{A_t L_t}, \qquad \tilde{c}_t = \frac{C_t}{A_t L_t}$$

Balanced growth path corresponds to steady values of normalized variables.

Why CRRA is Central for Balanced Growth

Chapter 4 emphasizes a sharp characterization:

- Exact balanced growth with constant interest in these models requires power utility (CRRA/log limit).
- Otherwise, saving behavior drifts systematically with growth in scale.

Practical Implication

Most quantitative macro models use CRRA not only for convenience, but for consistency with long-run growth facts.

Dynamic Programming Idea

Instead of choosing full sequences directly, define the value of starting from state x:

$$V(x) = \max_{\{x_{t+1}\}} \sum_{t=0}^{\infty} \beta^t \mathcal{F}(x_t, x_{t+1})$$

Stationarity allows decomposition into today + continuation value.

Appendix

Bellman Equation (4.22)

$$V(x) = \max_{x' \in \Gamma(x)} \{\mathcal{F}(x, x') + \beta V(x')\}$$

Policy function:

$$g(x) = \arg\max_{x' \in \Gamma(x)} \{\mathcal{F}(x, x') + \beta V(x')\}$$

- V: value of being in state x.
- g: optimal one-step-ahead decision rule.

Standard NGM in Recursive Form (P4)

$$V(k) = \max_{k' \in \Gamma(k)} \{u\left(f(k) + (1-\delta)k - k'\right) + \beta V(k')\}$$

with

$$\Gamma(k) = [0, f(k) + (1-\delta)k]$$

Current state: k. Control: k' (equivalently consumption).

How to Identify State Variables

A variable should be a state if it is:

- Predetermined at decision time.
- Payoff relevant for today and future.
- Not redundant given other states.

Common mistake

Leaving out a relevant predetermined variable breaks stationarity and yields wrong policy rules.

Periodic Productivity Example (Chapter 4.4.2)

If technology alternates deterministically A_h, A_l, recursive form uses two value functions:

$$V_h(k) = \max_{k' \in \Gamma_h(k)} \{u(A_h f(k) + (1-\delta)k - k') + \beta V_l(k')\}$$

$$V_l(k) = \max_{k' \in \Gamma_l(k)} \{u(A_l f(k) + (1-\delta)k - k') + \beta V_h(k')\}$$

State must include both k and the regime indicator.

Delayed Depreciation Example (Chapter 4.4.2)

If capital equals two lagged investments, $k_t = i_{t-1} + i_{t-2}$, then two states are needed.

Recursive representation uses $V(i', i)$ and choice i''.

Intuition

Non-standard technology often increases state dimension.

Core Properties of the Value Function

Under standard assumptions:

1. Bellman operator is a contraction ($\beta < 1$).
2. V is increasing when primitives are monotone.
3. V is concave when primitives/feasible set are concave-convex.
4. Interior points allow differentiability of V.
5. Policy function g is increasing.

Value Iteration Algorithm

Start from any bounded continuous V_0 (often $V_0 = 0$), then iterate:

$$V_{n+1}(x) = \max_{x' \in \Gamma(x)} \{\mathcal{F}(x, x') + \beta V_n(x')\}$$

Error shrinks geometrically in sup norm:

$$\|V_{n+1} - V\| \leq \beta \|V_n - V\|$$

Computational message: globally reliable convergence.

Appendix

Example 4.7: Recursive Solution in Log-Cobb-Douglas Case

With $u(c) = \log c$, $f(k) = Ak^{\alpha}$, $\delta = 1$:

$$V(k) = \max_{k' \geq 0} \{\log(Ak^{\alpha} - k') + \beta V(k')\}$$

Value iteration reveals the policy converges to

$$k' = \alpha\beta Ak^{\alpha}$$

Same rule obtained from sequential approach.

Functional Euler Equation

Bellman FOC (interior):

$$-\mathcal{F}_2(x, x') = \beta V'(x')$$

Envelope condition:

$$V'(x) = \mathcal{F}_1(x, g(x))$$

Combining:

$$-\mathcal{F}_2(x, g(x)) = \beta \mathcal{F}_1(g(x), g(g(x)))$$

This is the functional Euler equation.

Appendix

Why Functional Euler Is Useful but Not Enough

- It is a necessary condition for optimal policy.
- It can admit multiple candidate functions.
- Only one candidate attains the Bellman maximum globally.

Practice

Use functional Euler for intuition/local approximations, but verify with Bellman optimality.

Policy Dynamics in the Optimizing NGM

Given $k_{t+1} = g(k_t)$, dynamics are a first-order system in the state.

Steady states solve

$$\bar{k} = g(\bar{k})$$

Local approximation around steady state:

$$g(k) \approx a_0 + a_1(k - \bar{k})$$

Convergence speed is governed by slope a_1.

Stable vs Unstable Local Roots

Chapter 4.4.6 shows a second-order condition can yield two local roots for slope.

- One root satisfies $0 < a_1 < 1$: stable, monotone convergence.
- Another root exceeds 1: unstable, violates optimality/Bellman maximum.

Economic Selection

The Bellman problem chooses the stable policy branch.

Calibration Insight from Local Slope

Define curvature term near steady state (notation of chapter):

$$\Theta \equiv \beta^2 \frac{u' f''}{u''} > 0$$

Then local slope a_1 solves a quadratic expression.

Interpretation:

- $a_1 \approx 0$: very fast convergence.
- $a_1 \approx 1$: very slow convergence.

This replaces fixed-s Solow speed with preference-dependent dynamics.

Solow vs Optimizing NGM: Key Dynamic Difference

Solow

- s exogenous constant.
- Dynamics from technology + depreciation + fixed saving rule.

Same long-run logic, richer transition mechanism.

Optimizing NGM

- $s_t = s(k_t)$ endogenous.
- Dynamics shaped by u, f, and expectations.

What Chapter 4 Delivers

1. A unified language for dynamic choices in macroeconomics.
2. Euler equations as central intertemporal optimality conditions.
3. Clear distinction between finite and infinite horizon boundary logic.
4. nPg and TVC as fundamental infinite-horizon restrictions.
5. Recursive methods that scale to quantitative and stochastic models.

Takeaways for Research Practice

- Always verify well-posedness before solving.
- Use sequential methods for theory and derivations.
- Use recursive methods for computation and policy functions.
- Check TVC/nPg consistency when proposing candidate paths.
- Keep growth consistency in mind: CRRA is structural, not cosmetic.

Next Lecture (Chapter 5)

We move from individual optimization to dynamic equilibrium:

- Firms, households, and price determination.
- Decentralization of planner allocations.
- Competitive equilibrium as a dynamic system.

A1. From Generic Constraint to Sequence-Only Problem

If $x_{t+1} = h(x_t, y_t)$ can be inverted to $y_t = \hat{h}(x_t, x_{t+1})$, define

$$\mathcal{F}(x_t, x_{t+1}) \equiv \hat{\mathcal{F}}\big(\hat{h}(x_t, x_{t+1})\big)$$

Then (P1) becomes

$$\max_{\{x_{t+1}\}} \sum_{t=0}^{T} \beta^t \mathcal{F}(x_t, x_{t+1})$$

with $x_{t+1} \in \Gamma(x_t)$.
This is the standard form used for Euler equations in Chapter 4.

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A2. Two-Period Euler Equation with Kuhn-Tucker

FOCs:

$$u'(c_0) - \mu - \lambda = 0, \qquad \beta u'(c_1) - \frac{\mu}{1+r} = 0$$

If interior ($\lambda = 0$):

$$\mu = u'(c_0), \qquad \mu = \beta(1+r)u'(c_1)$$

Hence

$$u'(c_0) = \beta(1+r)u'(c_1)$$

Back

A3. Finite-Horizon NGM Euler Equation

From

$$\mathcal{L} = \sum_{t=0}^{T} \beta^t \left[u(c_t) + \mu_t k_{t+1} \right]$$

where $c_t = f(k_t) + (1-\delta)k_t - k_{t+1}$, FOC for k_{t+1} gives

$$-u'(c_t) + \beta u'(c_{t+1}) \left[f'(k_{t+1}) + 1 - \delta \right] + \mu_t = 0$$

At interior dates, $\mu_t = 0$.

Back

A4. EIS Under CRRA

Euler with gross return $R_{t,t+s}$:

$$u'(c_t) = \beta^s u'(c_{t+s}) R_{t,t+s}$$

CRRA derivative: $u'(c) = c^{-\sigma}$, so

$$\frac{c_{t+s}}{c_t} = (\beta^s R_{t,t+s})^{1/\sigma}$$

Take logs and differentiate w.r.t. $\log R_{t,t+s}$:

$$EIS = \frac{1}{\sigma}$$

Back

A5. Consolidating Budget Constraints with nPg

Start from

$$c_t + a_{t+1} = (1+r)a_t$$

At $t = 0$: $a_1 = (1+r)a_0 - c_0$.
At $t = 1$: $a_2 = (1+r)a_1 - c_1 = (1+r)^2 a_0 - (1+r)c_0 - c_1$. Continuing forward, for any $T \geq 0$:

$$a_{T+1} = (1+r)^{T+1} a_0 - \sum_{t=0}^{T} (1+r)^{T-t} c_t$$

Divide by $(1+r)^T$:

$$\sum_{t=0}^{T} \frac{c_t}{(1+r)^t} = a_0(1+r) - \frac{a_{T+1}}{(1+r)^T}$$

Now let $T \to \infty$ and impose nPg:

$$\lim_{T\to\infty} \frac{a_{T+1}}{(1+r)^T} \geq 0$$

Hence the lifetime budget set is

$$\sum_{t=0}^{\infty} \frac{c_t}{(1+r)^t} \leq a_0(1+r)$$

A6. Proposition 4.4 Logic (Sketch)

Given concavity, for any feasible comparison path,

$$\mathcal{F}(x_t, x_{t+1}) - \mathcal{F}(x_t^*, x_{t+1}^*)$$

is bounded above by tangent terms.

Summing and applying Euler equations collapses interior terms.

Remaining boundary term vanishes by TVC, implying candidate path gives weakly higher value than any feasible alternative.

Back

A7. Bellman Equation Derivation

Start from

$$V(x_0) = \max_{\{x_{t+1}\}} \sum_{t=0}^{\infty} \beta^t \mathcal{F}(x_t, x_{t+1})$$

Split first period:

$$V(x_0) = \max_{x_1 \in \Gamma(x_0)} \left\{ \mathcal{F}(x_0, x_1) + \beta \max_{\{x_{t+2}\}} \sum_{t=0}^{\infty} \beta^t \mathcal{F}(x_{t+1}, x_{t+2}) \right\}$$

Bracketed term is $V(x_1)$ by stationarity.

Back

A8. Why Contraction Implies Reliable Iteration

Bellman operator $\mathcal{T}$ satisfies

$$\|\mathcal{T}V - \mathcal{T}W\| \leq \beta\|V - W\|, \qquad \beta < 1$$

Banach fixed-point theorem implies:

- Unique fixed point V.
- Iteration $V_{n+1} = \mathcal{T}V_n$ converges to V from any V_0.
- Geometric convergence rate β.

Back

A9. Functional Euler from Bellman + Envelope

Bellman FOC at interior optimum $x' = g(x)$:

$$\mathcal{F}_2(x, g(x)) + \beta V'(g(x)) = 0$$

Envelope:

$$V'(x) = \mathcal{F}_1(x, g(x))$$

Substitute envelope one step ahead:

$$-\mathcal{F}_2(x, g(x)) = \beta \mathcal{F}_1(g(x), g(g(x)))$$

Back

A10. Finite-Horizon Closed Form (Example 4.1)

With log utility, Cobb-Douglas, and $\delta = 1$, Euler gives recursion:

$$\frac{1}{Ak_t^\alpha - k_{t+1}} = \beta \frac{\alpha A k_{t+1}^{\alpha-1}}{Ak_{t+1}^\alpha - k_{t+2}}$$

Using terminal condition $k_{T+1} = 0$ and backward induction yields

$$k_{t+1} = \alpha\beta \frac{1 - (\alpha\beta)^{T-t}}{1 - (\alpha\beta)^{T-t+1}} A k_t^\alpha$$

Back

A11. Infinite-Horizon Log Consumption-Saving (Example 4.3)

FOCs under consolidated budget:

$$\beta^t \frac{1}{c_t} = \lambda \left(\frac{1}{1+r} \right)^t$$

Thus

$$c_t = [\beta(1+r)]^t c_0$$

Plug into budget:

$$\sum_{t=0}^{\infty} \beta^t c_0 = a_0(1+r)$$

Hence $c_0 = a_0(1-\beta)(1+r)$.

Back

A12. Solow Link in Log-Cobb-Douglas NGM

From Euler and guess $k' = sAk^{\alpha}$ with $\delta = 1$:

$$\frac{1}{Ak^{\alpha} - k'} = \beta \frac{\alpha A (k')^{\alpha - 1}}{Ak'^{\alpha} - k''}$$

Consistency of the guess implies

$$s = \alpha\beta$$

so policy is $k' = \alpha\beta Ak^{\alpha}$.

Back

A13. Steady State Condition in Stationary NGM

At steady state, $k_{t+1} = k_t = \bar{k}$ and $c_{t+1} = c_t$. Euler simplifies to

$$1 = \beta \left[f'(\bar{k}) + 1 - \delta \right]$$

Then resources imply

$$\bar{c} = f(\bar{k}) - \delta \bar{k}$$

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A14. nPg vs TVC in One Line

- nPg: excludes paths with exploding debt in present value.
- TVC: excludes paths with leftover positive asset value at infinity.

Feasibility + optimality jointly select economically meaningful paths in infinite horizon.

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